Gonzalo Josue Spina

Buenos Aires, Argentina
LinkedIn | GitHub

PROFILE

Software engineer with 10 years of experience building distributed systems, currently working on a large-scale search engine and ML infrastructure at Mercado Libre. Completing a Data Science degree at the University of Buenos Aires; seeking Research Engineer roles in neurotechnology and AI.

EDUCATION

University of Buenos Aires - Faculty of Exact and Natural Sciences, Argentina *2021-Present*
Licenciatura en Ciencia de Datos (Data Science)
Two courses remaining; **average grade 8.44/10.**

Machine Learning: Machine learning; deep learning; natural language processing; computational statistics; reinforcement learning.
Mathematics: Real analysis and measure theory; probability; stochastic calculus / SDEs; optimal control.
Algorithms & Data Structures: Graph theory; graph traversal (DFS/BFS); dynamic programming; linear programming.
Neuroscience: Coursework in neural circuits and behavior, including electrophysiology and neurophysiology fundamentals.

EXPERIENCE

Sr. Software Engineer / Tech Lead - Search Infrastructure *Argentina*
Mercado Libre *Jul 2025 - Present*

- Lead engineering initiatives within a five-person team building a distributed search engine serving approximately **600K requests/minute** across Kubernetes clusters.
- Designed and implemented streaming aggregation components for organic and advertising retrieval, including a component, which filters and combines lexical and semantic advertising candidates. Built priority-aware asynchronous fan-in/fan-out using producer/consumer workers, futures and queues while preserving retrieval precedence and achieving approximately **20 ms p99 request latency in production**.
- **Built distributed A/B experiment validation across search stages**, counting an experiment as valid only when all participating components apply their assigned variants and satisfy the experiment's conditions.
- Led offline evaluation of **five embedding models** on a golden dataset, selecting a **4B-parameter Qwen model**. Built ETL pipelines on Vertex AI to re-embed and index catalogs totaling approximately **32 TB** across Brazil, Mexico and Argentina, and process approximately **10M daily catalog mutations**.
- Led the online A/B rollout of the selected Qwen 4B embedding model for semantic retrieval, including pre-population of hundreds of GB of Cassandra query-embedding cache while maintaining approximately **200 ms p95 semantic retrieval latency**.

SELECTED RESEARCH & ENGINEERING PROJECTS

EMG-Based Continuous Movement Decoding
Research engineering project · Project page *Jun 2026*

- Built an Arduino-based surface-EMG acquisition setup and collected original forearm EMG and movement data for continuous index-finger decoding.
- Designed and trained a causal temporal convolutional network (TCN) for online inference, achieving $R^2 = 0.688$ and Pearson correlation $r = 0.830$ across movement tasks.
- Developed causal signal-processing and temporal-alignment workflows for time-series data used in offline training and online inference.

Score-Based Generative Models & Stochastic Differential Equations
Academic monograph and poster · Project page *Jun 2026*
- Wrote a monograph and poster on score-based generative models via Itô diffusions, Kolmogorov equations, infinitesimal generators and reverse-time SDEs.

Instruction Tuning of a 7B Language Model
Natural Language Processing course project *Dec 2025*
- Fine-tuned a Mistral 7B language model with LoRA-based instruction tuning to generate sarcastic responses.

Predictive Modeling of Neural Population Activity
Academic project · Project page *Nov 2025*
- Built Poisson GLMs for mouse stimulus-reward recordings using binary stimulus time series, target-neuron history and population-history covariates; evaluated predictive performance using held-out log-likelihood.
- Found that stimulus identity improved held-out prediction and that population history added information in specific task conditions; temporal analysis did not establish a causal inhibitory link.

EARLIER EXPERIENCE

Sr Software Engineer *United States (Remote)*
Wagr / Yahoo Sports (acquired by Yahoo in 2023) *2022-2024*
- Built TypeScript/Node.js betting services and an online Bayesian recommender that updated user preferences from new interactions. Explored dynamic odds using linear-to-logistic saturation to model loss exposure.

Software Architect / Backend Lead - Founding Engineer *Switzerland (Remote)*
Sigi *2020-2022*
- Led three backend engineers in building a mental-health CRM from scratch, coordinating with two frontend teams and integrating calendar and remote-session services.

Software Engineer *Argentina*
Brubank *2019-2020*
- Owned critical banking systems covering central bank requirements and credit-bureau reporting; implemented critical regulatory changes, including foreign-exchange limits and required customer declarations.
- Planned and executed the migration from a banking monolith to a distributed, event-driven system on Kubernetes, moving two critical services built with Go, Node.js and Python.

SELECTED CONSULTING WORK

Applied Analytics & AI Tooling
Marketing agency projects
- Built a Go backend with n8n integrating Google Ads and Meta Ads APIs; computed trends over multiple rolling windows and top-k ad creative rankings, exposed through MCP for daily LLM-assisted campaign analysis.
- Derived socioeconomic and educational proxies with PCA and built exploratory regional scores using brand penetration, target-age overlap and word-of-mouth assumptions. Findings informed three private health coverage launch proposals, alongside client discussions and the agency's commercial assessment.

TECHNICAL SKILLS

Languages: Python, Go, C++, SQL
ML & Statistics: Semantic retrieval, embedding evaluation, instruction tuning, Poisson GLMs, TCNs, PCA, A/B experimentation
Neural Signals: EMG acquisition, filtering, RMS / envelopes, temporal alignment, neural encoding / decoding
Systems & Data: Distributed streaming, asynchronous concurrency, ETL, Kubernetes, Docker, Vertex AI, PostgreSQL, Redis, Datadog, MCP, n8n